



Heat exchanger design model

Assumptions

- Operation is steady state, and there is no phase change.
- Fluid properties are constant.
- No heat is lost to the surroundings.
- Each hairpin module contains two straight legs.
- Heat transfer area is based on the outside surface of the inner pipe.
- The bent regions of the hairpin are excluded from the heat transfer area and the distributed friction length.

Heat exchanger geometry

The subscripts i and a identify the inner pipe and annulus, respectively. Assume one straight leg is 6 meters long.

1. Total straight leg length (meters):

$$l_{\text{straight,total}} = 2nl_{\text{leg}}$$

where n is the number of hairpin modules, and l_{leg} is the length of one straight leg in meters.

2. Inner pipe flow area (square meters):

$$a_i = \frac{\pi d_{\text{inner,in}}^2}{4}$$

where $d_{\text{inner,in}}$ is the inside diameter of the inner pipe in meters.

3. Annulus flow area (square meters):

$$a_a = \frac{\pi(d_{\text{outer,in}}^2 - d_{\text{inner,out}}^2)}{4}$$

where $d_{\text{outer,in}}$ is the inside diameter of the outer pipe in meters, and $d_{\text{inner,out}}$ is the outside diameter of the inner pipe in meters.

4. Annulus hydraulic diameter (meters):

$$d_{h,a} = d_{\text{outer,in}} - d_{\text{inner,out}}$$

5. Heat transfer area (square meters):

$$a_o = \pi d_{\text{inner,out}} l_{\text{straight,total}}$$



Flow calculations

Calculate the flow quantities separately for each passage using the properties and mass flow rate of the fluid in that passage. Assume the following:

- densities of 1040 for the process fluid and 975 for the hot water, in kilograms per cubic meter.
- dynamic viscosities of 0.00164 for the process fluid and 0.00038 for the hot water, in pascal seconds.
- specific heat capacities of 3600 for the process fluid and 4200 for the hot water, in joules per kilogram kelvin.
- thermal conductivities of 0.44 for the process fluid and 0.66 for the hot water, in watts per meter kelvin.

6. Flow velocity (meters per second):

$$v = \frac{\dot{m}}{\rho a}$$

where $\dot{m}$ is the mass flow rate in kilograms per second, ρ is the fluid density in kilograms per cubic meter, and a is the flow area in square meters. Use a_i for the inner pipe and a_a for the annulus.

7. Reynolds number (dimensionless):

$$\text{Re} = \frac{\rho v d_h}{\mu}$$

where d_h is the hydraulic diameter in meters, and μ is the dynamic viscosity in pascal seconds. Use $d_{\text{inner,in}}$ for the inner pipe and $d_{h,a}$ for the annulus.

8. Prandtl number (dimensionless):

$$\text{Pr} = \frac{c_p \mu}{k}$$

where c_p is the specific heat capacity in joules per kilogram kelvin, and k is the thermal conductivity in watts per meter kelvin.

Inner pipe heat transfer calculations

Use the laminar equations when $\text{Re}_i \leq 2300$, or the turbulent equations when $\text{Re}_i \geq 10000$. Both calculation paths require $0.5 \leq \text{Pr}_i \leq 2000$.

9. Graetz number for laminar flow (dimensionless):

$$\text{Gz}_i = \frac{\text{Re}_i \text{Pr}_i d_{\text{inner,in}}}{l_{\text{straight,total}}}$$



10. Nusselt number for laminar flow (dimensionless):

$$\text{Nu}_i = 3.66 + \frac{0.0668 \text{Gz}_i}{1 + 0.04 \text{Gz}_i^{2/3}}$$

11. Smooth pipe friction factor for turbulent heat transfer (dimensionless):

$$f_0 = \left[-1.8 \log_{10} \left(\frac{6.9}{\text{Re}_i} \right) \right]^{-2}$$

12. Nusselt number for turbulent flow (dimensionless):

$$\text{Nu}_i = \frac{(f_0/8)(\text{Re}_i - 1000)\text{Pr}_i}{1 + 12.7 \sqrt{f_0/8} (\text{Pr}_i^{2/3} - 1)}$$

13. Heat transfer coefficient (watts per square meter kelvin):

$$h_i = \frac{\text{Nu}_i k_i}{d_{\text{inner},\text{in}}}$$

Annulus heat transfer calculations

Use the laminar equation when $\text{Re}_a \leq 2300$, or the turbulent equations when $\text{Re}_a \geq 10000$. Both calculation paths require $0.5 \leq \text{Pr}_a \leq 2000$.

14. Diameter ratio (dimensionless):

$$\alpha = \frac{d_{\text{inner},\text{out}}}{d_{\text{outer},\text{in}}}$$

15. Nusselt number for laminar flow (dimensionless):

$$\text{Nu}_a = \text{Nu}_{a,1} + \frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} (\text{Nu}_{a,2} - \text{Nu}_{a,1})$$

where the subscripts 1 and 2 identify the adjacent table entries that bracket α . Use the laminar annulus interpolation only for $0.05 \leq \alpha < 1$.

α	0.05	0.10	0.25	0.50	1.00
Nu_a	17.46	11.56	7.37	5.74	4.86

16. Modified Reynolds number for turbulent flow (dimensionless):

$$\text{Re}_a^* = \text{Re}_a \frac{(1 + \alpha^2) \ln(\alpha) + (1 - \alpha^2)}{(1 - \alpha)^2 \ln(\alpha)}$$

17. Friction factor for turbulent heat transfer (dimensionless):

$$f_a = [1.8 \log_{10}(\text{Re}_a^*) - 1.5]^{-2}$$



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18. Correction term (dimensionless):

$$k_1 = 1.07 + \frac{900}{\text{Re}_a} - \frac{0.63}{1 + 10\text{Pr}_a}$$

19. Geometry factor (dimensionless):

$$f_{\text{geom}} = 0.75\alpha^{-0.17}$$

20. Length correction (dimensionless):

$$f_{\text{length}} = 1 + \left(\frac{d_{h,a}}{l_{\text{straight,total}}} \right)^{2/3}$$

21. Nusselt number for turbulent flow (dimensionless):

$$\text{Nu}_a = \frac{(f_a/8)\text{Re}_a\text{Pr}_a}{k_1 + 12.7\sqrt{f_a/8}(\text{Pr}_a^{2/3} - 1)} f_{\text{length}} f_{\text{geom}}$$

22. Heat transfer coefficient (watts per square meter kelvin):

$$h_a = \frac{\text{Nu}_a k_a}{d_{h,a}}$$

Thermal resistance and overall heat transfer coefficient

Assume the following:

- a fouling resistance of 0.0002 for the process fluid and 0.0001 for the hot water, in square meters kelvin per watt.
- pipe wall thermal conductivity of 45 watts per meter kelvin.

23. Inner pipe convection resistance (square meters kelvin per watt):

$$r_{\text{conv},i} = \frac{d_{\text{inner,out}}}{d_{\text{inner,in}} h_i}$$

24. Inner pipe fouling resistance (square meters kelvin per watt):

$$r_{\text{foul},i} = \frac{d_{\text{inner,out}} r_{f,i}}{d_{\text{inner,in}}}$$

where $r_{f,i}$ is the fouling resistance based on the inside surface of the inner pipe, in square meters kelvin per watt.



25. Pipe wall conduction resistance (square meters kelvin per watt):

$$r_{\text{wall}} = \frac{d_{\text{inner,out}} \ln(d_{\text{inner,out}}/d_{\text{inner,in}})}{2k_s}$$

where k_s is the pipe wall thermal conductivity in watts per meter kelvin.

26. Annulus fouling resistance (square meters kelvin per watt):

$$r_{\text{foul},a} = r_{f,a}$$

where $r_{f,a}$ is the fouling resistance on the outside surface of the inner pipe, in square meters kelvin per watt.

27. Annulus convection resistance (square meters kelvin per watt):

$$r_{\text{conv},a} = \frac{1}{h_a}$$

28. Total thermal resistance (square meters kelvin per watt):

$$r_{\text{total}} = r_{\text{conv},i} + r_{\text{foul},i} + r_{\text{wall}} + r_{\text{foul},a} + r_{\text{conv},a}$$

29. Overall heat transfer coefficient (watts per square meter kelvin):

$$u_o = \frac{1}{r_{\text{total}}}$$

Thermal performance calculations

The subscripts p and w identify the process fluid and hot water, respectively.

30. Heat capacity rate (watts per kelvin):

$$C = \dot{m}c_p$$

31. Minimum heat capacity rate (watts per kelvin):

$$C_{\text{min}} = \min(C_p, C_w)$$

32. Maximum heat capacity rate (watts per kelvin):

$$C_{\text{max}} = \max(C_p, C_w)$$

33. Heat capacity ratio (dimensionless):

$$C_r = \frac{C_{\text{min}}}{C_{\text{max}}}$$

34. Number of transfer units (dimensionless):

$$\text{NTU} = \frac{u_o a_o}{C_{\text{min}}}$$



35. Effectiveness for parallel flow (dimensionless):

$$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r}$$

36. Effectiveness for countercurrent flow (dimensionless):

$$\varepsilon = \begin{cases} \frac{NTU}{1 + NTU} & C_r = 1 \\ \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \exp[-NTU(1 - C_r)]} & C_r \neq 1 \end{cases}$$

37. Heat transfer rate (watts):

$$q = \varepsilon C_{\min}(t_{w,in} - t_{p,in})$$

where $t_{w,in}$ is the hot water inlet temperature, and $t_{p,in}$ is the process fluid inlet temperature, both in degrees Celsius. The subscript **in** identifies an inlet condition.

38. Process fluid outlet temperature (degrees Celsius):

$$t_{p,out} = t_{p,in} + \frac{q}{C_p}$$

The subscript **out** identifies an outlet condition.

39. Hot water outlet temperature (degrees Celsius):

$$t_{w,out} = t_{w,in} - \frac{q}{C_w}$$

40. Minimum approach temperature for parallel flow (degrees Celsius):

$$\Delta t_{\min} = \min(t_{w,in} - t_{p,in}, t_{w,out} - t_{p,out})$$

41. Minimum approach temperature for countercurrent flow (degrees Celsius):

$$\Delta t_{\min} = \min(t_{w,in} - t_{p,out}, t_{w,out} - t_{p,in})$$

Pressure drop calculations

The subscript **d** identifies the Darcy friction factor. Calculate the pressure drop separately for each flow passage.

42. Darcy friction factor for laminar flow (dimensionless):

$$f_{d,i} = \frac{64}{Re_i}$$
$$f_{d,a} = \frac{64}{Re_a} \frac{(1 - \alpha)^2}{1 + \alpha^2 + \frac{1 - \alpha^2}{\ln \alpha}}$$



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43. Darcy friction factor for turbulent flow (dimensionless):

$$f_d = \left[-1.8 \log_{10} \left(\left(\frac{\epsilon_r}{3.7d_h} \right)^{1.11} + \frac{6.9}{\text{Re}} \right) \right]^{-2}$$

where ϵ_r is the absolute roughness in meters. Assume an absolute roughness of 0.000045 meters for both flow passages.

44. Total minor loss coefficient (dimensionless):

$$k_{\text{total}} = 1.5(1 + n) + 2 \times \max(0, n - 1)$$

45. Pressure drop (pascals):

$$\Delta p = \left(f_d \frac{l_{\text{straight, total}}}{d_h} + k_{\text{total}} \right) \frac{\rho v^2}{2}$$

Cost calculations

Calculate the electrical power and annual pumping cost separately for each pump.

Assume the following:

- overall pump efficiency of 0.7 for both pumps.
- an annual operating time of 8000 hours per year.
- an electricity price of 0.12 Canadian dollars per kilowatt hour.

46. Annualised capital cost (Canadian dollars per year):

$$c_{\text{capital, annual}} = 0.18nc_{\text{module}}$$

where c_{module} is the installed cost of one module in Canadian dollars.

47. Pump electrical power (watts):

$$p_{\text{electric}} = \frac{\Delta p(\dot{m}/\rho)}{\eta_{\text{pump}}}$$

where η_{pump} is the dimensionless overall pump efficiency.

48. Annual pumping cost (Canadian dollars per year):

$$c_{\text{pump}} = \frac{p_{\text{electric}}}{1000} h_{\text{year}} e_{\text{price}}$$

where h_{year} is the annual operating time in hours per year, and e_{price} is the electricity price in Canadian dollars per kilowatt hour.

49. Total annual cost (Canadian dollars per year):

$$c_{\text{annual}} = c_{\text{capital, annual}} + c_{\text{pump, p}} + c_{\text{pump, w}}$$